

N-Queens Problem

PROGRAM:

```
def is_safe(board, row, col, n):
    for i in range(row):
        if board[i] == col or \
            abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve_n_queens(board, row, n):
    if row == n:
        print_board(board, n)
        return True
    for col in range(n):
        if is_safe(board, row, col, n):
            board[row] = col
            if solve_n_queens(board, row + 1, n):
                return True
    return False

def print_board(board, n):
    for i in range(n):
        line = ["Q" if board[i] == j else "." for j in range(n)]
        print(" ".join(line))
    print()

n = int(input("enter the value for n : "))
board = [-1] * n
solve_n_queens(board, 0, n)
```

OUTPUT:

```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb  3 2026, 18:22:25) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:\New folder (9)\queen.py =====
enter the value for n : 4
. Q . .
. . . Q
Q . . .
. . Q .

>>>
===== RESTART: D:\New folder (9)\queen.py =====
enter the value for n : 8
Q . . . . .
. . . . Q . .
. . . . . Q
. . . . Q . .
. . Q . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . .

>>> |
```